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Structural and electronic properties of Ga_2O_3-Al_2O_3 alloys

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Ga_2O_3 is emerging as an important electronic material. Alloying with Al_2O_3 is a viable method to achieve carrier confinement, to increase the bandgap, or to modify the lattice parameters. However, the two materials have very different ground-state crystal structures (monoclinic β-gallia for Ga_2O_3 and corundum for Al_2O_3). Here, we use hybrid density functional theory calculations to assess the alloy stabilities and electronic properties of the alloys. We find that the monoclinic phase is the preferred structure for up to 71% Al incorporation, in close agreement with experimental phase diagrams, and that the ordered monoclinic $AlGaO_3$ alloy is exceptionally stable. We also discuss bandgap bowing, lattice constants, and band offsets that can guide future synthesis and device design efforts. *Published by AIP Publishing*. https://doi.org/10.1063/1.5036991

Ga_2O_3 is a wide-bandgap semiconductor with a fundamental bandgap of 4.76 eV.[1] Its transparency into the UV allows applications such as solar-blind photodetectors[2] and deep-UV transparent contacts.[3] Its high breakdown voltage and low effective mass and the ability to achieve controllable *n*-type doping[4–6] render it suitable for high-voltage field effect transistors[7,8] and Schottky barrier diodes.[9] *p*-Type doping is unlikely to occur.[10,11] These properties, along with the availability of high-quality single-crystal substrates, render Ga_2O_3 a promising material for next-generation devices.

Many applications require a material with a bandgap even larger than Ga_2O_3, as a dielectric, or to form heterostructures for carrier confinement, or to enable optoelectronics deeper into the UV.[12] Al_2O_3 is an excellent candidate, with a bandgap of 8.82 eV in the corundum (sapphire) phase.[13] Sapphire can also be used as a substrate for the growth of Ga_2O_3.[14–18] Alloy formation also permits tuning of the lattice parameters, which can minimize stress when the material is grown on a substrate and may offer additional control over the transport properties. Ga_2O_3-Al_2O_3 alloys [$(Al_xGa_{1-x})_2O_3$] have already been grown using chemical vapor deposition (CVD),[14–16] pulsed laser deposition (PLD),[17–20] or molecular beam epitaxy (MBE)[12,21–23] or through grinding or solid combustion synthesis.[24–28]

The ground-state crystal structures of Ga_2O_3 and Al_2O_3 are very different: Ga_2O_3 adopts the monoclinic (β) phase with symmetry group $C2/m$, while Al_2O_3 adopts the corundum (α) phase with symmetry group $R\bar{3}c$. Corundum Ga_2O_3 is also known as α-Ga_2O_3, while the θ phase of Al_2O_3 has the same structure as the monoclinic β phase of Ga_2O_3. The primitive cells of both structures contain 10 atoms (see insets of Fig. 1). In the monoclinic phase, two types of cation environments are present: an octahedrally coordinated and a tetrahedrally coordinated site. In contrast, only octahedrally coordinated cation sites are present in the corundum phase. These different crystal structures and coordination environments raise the question what the preferred crystal structure will be in $(Al_xGa_{1-x})_2O_3$ alloys as a function of the alloy composition. Experimentally reported upper limits on the stability of the monoclinic structure have exhibited a wide range, with maximum Al concentrations ranging from 67 to 85%.[14,17–19,21,25–27] For the corundum structure, the upper limit is lower, with maximum Ga concentrations between 12 and 25%.[15,16,26,28]

There is clearly a need to elucidate the stability of $(Al_xGa_{1-x})_2O_3$ alloys in different structures and the evolution of the lattice parameters. In parallel, information about the change in bandgap as a function of the alloy composition is required for device applications; a strong nonlinearity ("bandgap bowing") is expected due to the structure mismatch between the binary endpoints. Information about band alignments between Ga_2O_3 and Al_2O_3 in their different phases is also currently incomplete, with experimental measurements reporting a wide spread of values.[29–32] In the

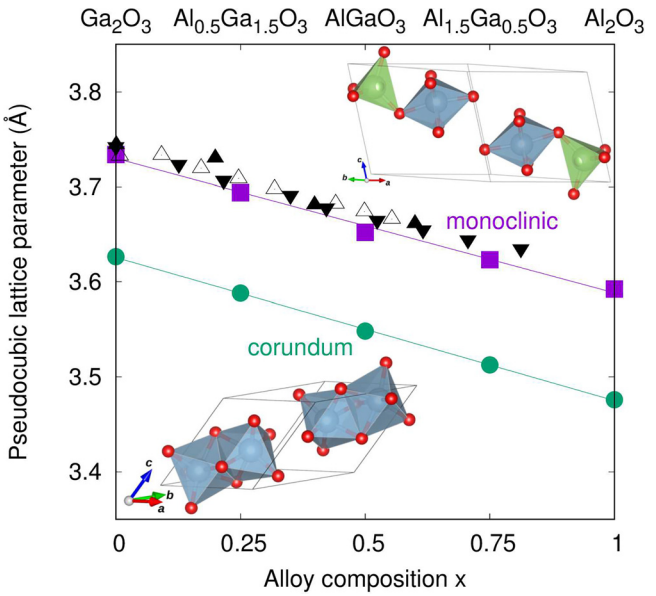


FIG. 1. Pseudocubic lattice parameter as a function of alloy composition x for the corundum and monoclinic structures. Experimental values for monoclinic are shown as solid black triangles (Ref. 24), open triangles (Ref. 19), and solid downward triangles (Ref. 25). The top inset shows the monoclinic unit cell and the bottom inset the corundum unit cell. Octahedrally coordinated sites are indicated in blue and tetrahedrally coordinated sites in green. Small red spheres depict O atoms.

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present work, we address all of these issues using hybrid density functional theory (DFT) calculations.

Our DFT calculations employ projector augmented wave (PAW) potentials[33] as implemented in the *Vienna Ab-initio Simulation Package* (VASP)[34] code, with a 500 eV cut-off in the plane-wave expansion and a $4\times4\times4$ **k**-point grid. All structures were considered converged when all residual forces were smaller than 5 meV/Å and all stress components were smaller than 50 meV/Å^3. Including the Ga 3*d* electrons in the valence is important; if they are treated as core electrons, the corundum α phase of Ga_2O_3 is 56 meV lower in energy than the monoclinic *β* phase, which disagrees with the fact that the monoclinic phase is the actual ground state of Ga_2O_3.[35] We use the common approach of adjusting the mixing parameter α in the hybrid functional of Heyd, Scuseria, and Ernzerhof (HSE)[36] to reproduce the experimental bandgap. A single value of α needs to be used in all calculations involving both Ga_2O_3 and Al_2O_3. As shown in Table I, $\alpha=0.32$ produces good values for the well-established gaps of *β*-Ga_2O_3 and α-Al_2O_3 and also yields structural parameters, which is in very good agreement with the experiment. We used 10-atom monoclinic and corundum cells and explicitly considered all alloys that can be modeled within these cells. Band alignments with respect to vacuum were obtained from surface calculations using 120-atom slabs terminated along the $(10\bar{1}0)$ direction for corundum and 100-atom slabs terminated along the (010) direction for the monoclinic structure. A 20-Å vacuum layer, which is sufficient to eliminate interactions between periodic images, was used together with a $2\times4\times1$ **k**-point sampling. Full relaxation of internal coordinates and in-plane lattice constants was performed. Both surface orientations are nonpolar in the sense that they contain both atom types in each layer.

The dependence of lattice parameters on the alloy composition is summarized in Fig. 1 by plotting the pseudocubic lattice parameter (the cube root of the volume per formula unit). The corundum structure has a smaller volume compared to the monoclinic structure because it is more compact due to all cations being octahedrally coordinated, while in the monoclinic structure, half the sites are tetrahedrally coordinated. When alloying Ga_2O_3 with Al, the lattice parameters shrink, consistent with the smaller atomic radius of Al compared to Ga. The change in the pseudocubic lattice parameter is linear, in agreement with Vegard's law. The same trend is observed in the experimental values for the monoclinic structure, also included in Fig. 1; the fact that the experimental data points lie slightly above the calculated curve is consistent with the small underestimate of our calculated lattice parameters compared to the experiment (Table I). Our results show that the individual lattice parameters (*a*, *b*, *c*, and *β*) closely follow Vegard's law, and the values for alloys can therefore be obtained by linear interpolation of the entries in Table I.

We examine the stability of alloys by comparing their enthalpies of formation (ΔH)

$$\Delta H\left[(Al_xGa_{1-x})_2O_3\right] = E\left[(Al_xGa_{1-x})_2O_3\right] - (1-x)\,E[Ga_2O_3] - x\,E[Al_2O_3], \quad (1)$$

where $E[Ga_2O_3]$ and $E[Al_2O_3]$ are the energies of the lowest energy structures, corundum for Al_2O_3 and monoclinic *β*-gallia for Ga_2O_3. Within a regular solution model, the enthalpy of mixing of heterostructural alloys is given by

$$\Delta H(x) = x(1-x)\Omega_0 + \Omega_1(x^2-x)(x-0.5) + (1-x)H_0 + xH_1, \quad (2)$$

where Ω_0 is the regular alloy interaction parameter, Ω_1 describes the asymmetry of the enthalpy of mixing, and H_0 and H_1 are the enthalpies of the endpoints. In Fig. 2, we plot the calculated $\Delta H(x)$ for the lowest-energy alloy structures using filled symbols at each composition, as well as a fit to Eq. (2). For corundum, we fit over the entire composition range, obtaining $\Omega_0=0.164$ eV and $\Omega_1=0.012$ eV. For the monoclinic phase, Eq. (2) fails to describe the mixing enthalpy due to the local minimum at $AlGaO_3$; instead, we perform separate fits over the intervals 0%–50% and 50%–100% Al without the asymmetry terms and with the

TABLE I. Calculated and experimental lattice parameters and bandgaps for the α (corundum) and *β* (monoclinic) phases of Ga_2O_3 and the α (corundum) and *θ* (monoclinic) phases of Al_2O_3.

	α-Ga_2O_3		*β*-Ga_2O_3		α-Al_2O_3		*θ*-Al_2O_3	
	Calc.	Exp.	Calc.	Exp.	Calc.	Exp.	Calc.	Exp.
a (Å)	4.97	4.98[a]	12.21	12.21[c]	4.74	4.76[e]	11.75	11.85[g]
b (Å)	4.97	4.98[a]	3.03	3.04[c]	4.74	4.76[e]	2.92	2.90[g]
c (Å)	13.39	13.43[a]	5.79	5.80[c]	12.94	12.99[e]	5.57	5.62[g]
β	…	…	103.80°	103.83°[c]	…	…	103.80°	103.83°[g]
E_{gap}^{direct} (eV)	5.59	5.32[b]	4.870	4.76[d]	8.82	8.8[f]	7.51	
$E_{gap}^{indirect}$ (eV)	5.34		4.865		…	…	7.24	

[a]Reference 37.
[b]Reference 14.
[c]Reference 38.
[d]Reference 1.
[e]Reference 39.
[f]Reference 13.
[g]Reference 40.

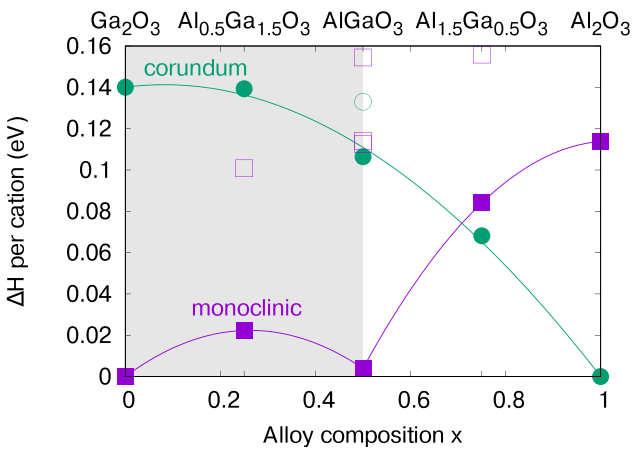


FIG. 2. Enthalpy of formation as a function of the alloy composition for the corundum and monoclinic structures. The shaded area indicates the composition range where Al preferentially occupies the octahedral sublattice in the monoclinic structure. The lines are fits as described in the text. The lowest energy structures are indicated by filled symbols and the higher-energy structures by open symbols.

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composition ranges scaled accordingly. This choice treats the octahedral and tetrahedral cations separately, where Al preferentially fills the octahedral sublattice for compositions up to 50%, as described below. These fits yield Ω_0 values of 0.081 eV and 0.101 eV for the two respective composition regimes; we thus find that the alloy interaction parameters are smaller in the monoclinic phase compared to the corundum phase.

Temperature effects are included in the Gibbs free energy of mixing, $\Delta G(x) = \Delta H(x) - T\Delta S(x)$, where the mixing entropy $\Delta S(x)$ is given by a random mixture model. Full miscibility, as determined by the common tangent method, is achieved at 952 K for corundum; for the monoclinic phase in the range up to $x = 0.5$, this is achieved at 472 K and for $x > 0.5$ at 584 K. These temperatures are an estimate of the upper bound since the model neglects vibrational entropy and additional configuration interactions.

Starting on the Ga_2O_3 side, the monoclinic crystal structure has the lowest energy; for bulk Ga_2O_3, the corundum phase is significantly higher in energy (by 0.14 eV per cation) than the monoclinic phase, consistent with the monoclinic phase being the stable structure.[35] In the alloys, the monoclinic phase remains energetically preferable for Al concentrations up to 71%. For a higher Al content, the corundum structure is preferred, and for pure Al_2O_3, the energy per cation site is 0.11 eV higher for monoclinic relative to the corundum ground state. For x values in the vicinity of 0.71, the energy difference between corundum and monoclinic is small and can easily be overcome at high temperatures, explaining why some experiments have observed the monoclinic structure for Al concentrations above 71%[17–19,21,25–27] or the corundum structure for Al concentrations lower than 71%.[14–16,26,28]

The energetics in the alloy strongly correlate with the preferred coordination environment of Al and Ga: Al atoms are found to always favor octahedral positions, which is the coordination of the cation sites in corundum. In the monoclinic structure, only half the sites are octahedrally coordinated, and in the alloys, we indeed find that these sites are preferentially occupied by Al atoms. This explains the relatively low ΔH for the Al content up to 50% Al (Fig. 2). Above 50%, Al is forced to incorporate on tetrahedral sites, significantly increasing ΔH of the monoclinic-phase alloys. The open symbols in Fig. 2 indicate higher-energy structures. For the monoclinic structures (squares), the higher energy reflects the energy cost for Al atoms to occupy tetrahedral positions. For corundum, the higher energy cost is associated with a different arrangement of the cations, where having a nearest neighbor of the same atom type leads to a larger energy. Details about the structure and energetics of the higher-energy structures are presented in the supplementary material.

At $x = 0.5$, the monoclinic $AlGaO_3$ structure is an ordered alloy with an exceptionally low enthalpy of formation (4 meV). In this structure, all Al atoms are on octahedral sites and all Ga atoms on tetrahedral sites. This type of structure was previously found to be highly stable for alloys of Ga_2O_3 with In_2O_3, where In preferentially incorporates on the octahedral sites.[41] Such an arrangement also minimizes the local strain since the local environment around both types of cations is very close to the local environment in their bulk ground-state structures.

We now examine the electronic structure of the alloys. Figure 3 shows the calculated bandgaps for the lowest-energy structures in both monoclinic and corundum phases. The bandgaps reported are fundamental bandgaps; the optical bandgaps can depend on the polarization of the light, especially in the monoclinic phases.[1,42–44] In the monoclinic phase, the valence-band maximum (VBM) of both Ga_2O_3 and Al_2O_3 is located on the I-L line in the Brillouin zone,[5] and the conduction-band minimum (CBM) is at the Γ point. This indirect nature of the gap is maintained throughout the alloy composition range, with the difference between direct and indirect bandgaps increasing from 5 meV for $x = 0$ to 270 meV for $x = 1$ and the VBM shifting toward the L point. In the corundum phase, the bandgap is direct near $x = 1$; adding Ga (i.e., decreasing x) leads to an indirect bandgap. In corundum Ga_2O_3, the direct bandgap is 250 meV above the indirect gap.

Bowing parameters b are obtained from

$$E_g(x) = (1-x)E_g[Ga_2O_3] + xE_g[Al_2O_3] - bx(1-x), \quad (3)$$

where $E_g[Ga_2O_3]$ and $E_g[Al_2O_3]$ are the gaps of the end points (Table I). For the monoclinic phase, we find $b = 1.78$ eV (indirect) and $b = 1.87$ eV (direct), and for corundum, $b = 0.93$ eV (indirect) and $b = 1.37$ eV (direct). Generally, good agreement is found between calculated and experimental values, with the largest deviations occurring for the high Al-limit of monoclinic alloys. We attribute this deviation to the lower crystalline quality of experimental samples as the monoclinic phase becomes unstable above $x = 0.71$ (Fig. 2).

Information about band alignments is essential for the design of heterostructures.[22,23,45,46] In Fig. 4(a), we summarize our results as determined by alignment to the vacuum level. The CBM of the Al_2O_3 compounds is located higher in energy compared to Ga_2O_3, consistent with the energy of the cation s states that dominate in the CBM. For a heterostructure in the monoclinic phase, the conduction-band offset between Ga_2O_3 and Al_2O_3 is 2.74 eV; in the corundum phase, it is 3.24 eV. The valence-band offsets are relatively small, as the VBM is dominated by O p states. Assuming

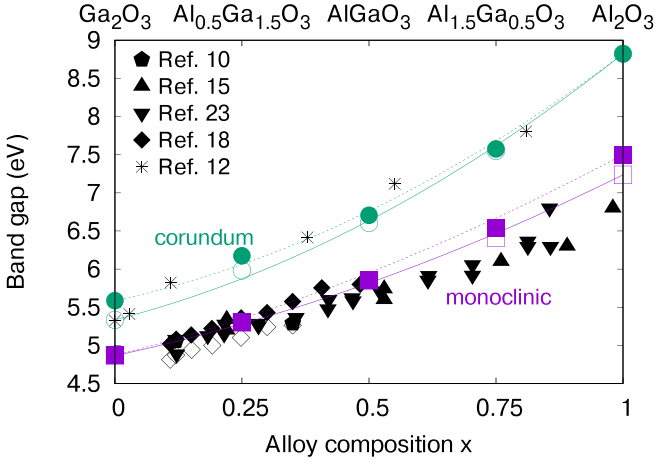


FIG. 3. Direct (solid symbols) and indirect (open symbols) bandgaps for the monoclinic and corundum structures as a function of the alloy composition. The solid (dashed) lines are quadratic fits to the indirect (direct) bandgaps, as described in the text. The computed values are shown in color and experimental values in black.

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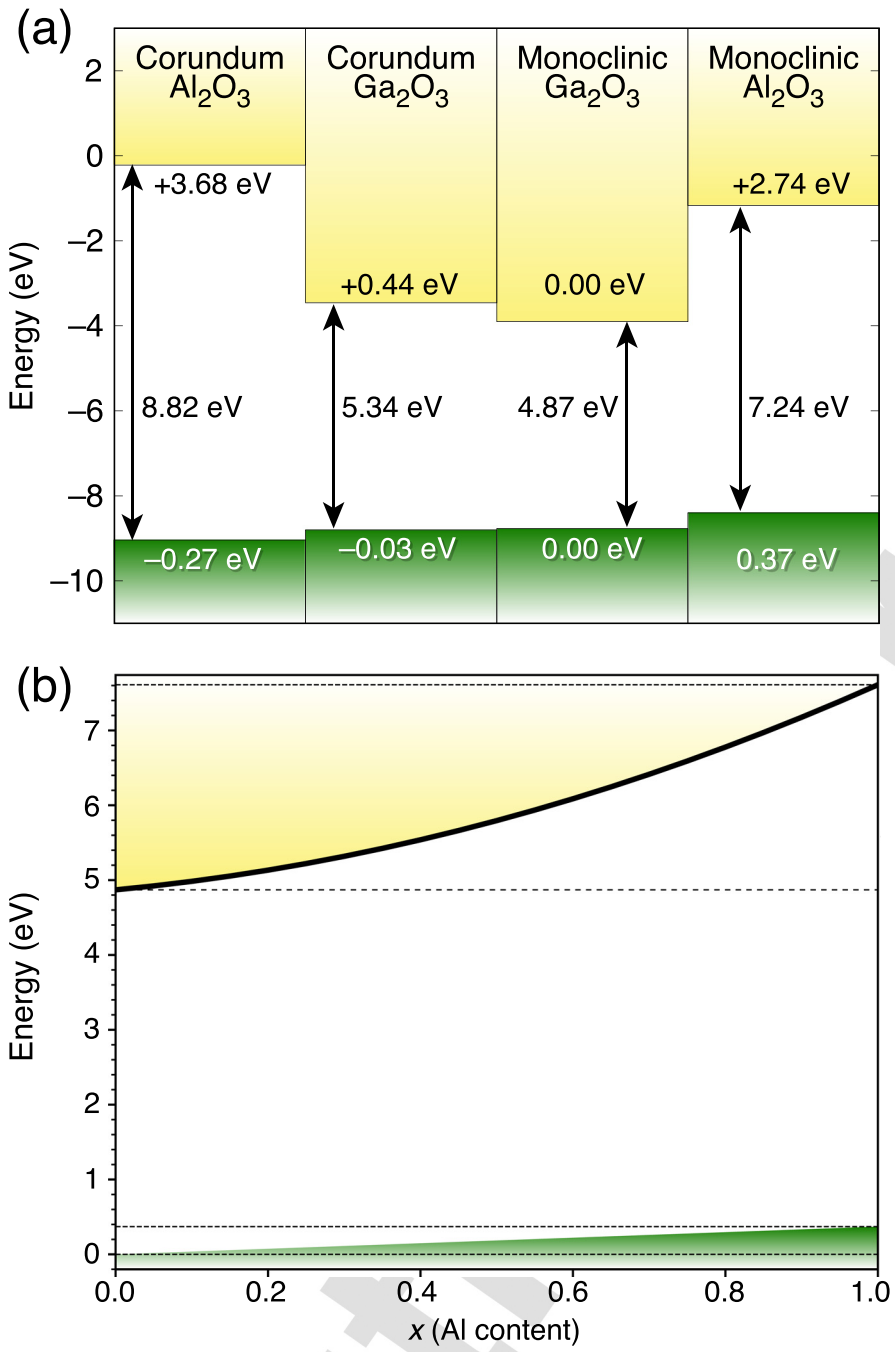


FIG. 4. (a) Valence- and conduction-band edges with respect to the vacuum level for monoclinic and corundum Ga_2O_3 and Al_2O_3. The values at the VBM indicate valence-band offsets and values at the CBM conduction-band offsets, all with respect to monoclinic Ga_2O_3 (valence-band edge at −8.77 eV and conduction-band edge at −3.90 eV). Bandgaps are also indicated. (b) Predicted band offsets as a function of the alloy concentration for alloys in the monoclinic structure.

that the absolute position of the VBM changes linearly, the bandgap information from Fig. 3 can be combined with the alignment from Fig. 4(a) to predict band offsets in monoclinic alloy heterostructures, as shown in Fig. 4(b). Experimentally, band offsets have been determined using X-ray photoelectron spectroscopy (XPS)[29–31] and C-V measurements.[29,32] The reported values exhibit a large spread, partly due to the fact that the overlayers are not always crystalline. Comparisons with our calculated values are therefore difficult.

In conclusion, we performed a detailed first-principles study of Ga_2O_3-Al_2O_3 alloys in both the corundum and monoclinic structures. We find that the pseudocubic lattice constant decreases linearly with the increasing Al concentration, while the bandgap increases but exhibits significant bowing. For alloy concentrations smaller than 71% Al, the monoclinic structure is the preferred structure, with Al preferentially occupying the octahedral positions. This preference for the octahedral site also explains the unusual stability of the ordered monoclinic $AlGaO_3$ alloy. We find that the technologically attractive monoclinic $(Al_xGa_{1-x})_2O_3$ alloys should be fully miscible over the composition range of 0%–50% at typical growth temperatures. Ga_2O_3 and Al_2O_3 have large conduction-band offsets, 2.74 eV for monoclinic and 3.24 eV for corundum structures. Band offsets in $(Al_xGa_{1-x})_2O_3$ alloys can be predicted by combining our calculated values for alloy bandgaps with interpolated values for band offsets between the binaries.

See supplementary material for details about the structure and energetics of the higher-energy structures.

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